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Data-driven heating and cooling load predictions for non-residential buildings based on support vector machine regression and NARX Recurrent Neural Network: A comparative study on district scale

D. Koschwitz*, J. Frisch, C. van Treeck

RWTH Aachen University, Institute of Energy Efficiency and Sustainable Building (E3D), Aachen, D-52074, Germany

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ABSTRACT

Predicting building energy consumption is essential for planning and managing energy systems. In recent times, numerous studies focus on load forecasting models dealing with a wide range of different methods. In addition to Artificial Neural Networks (ANN), especially Support Vector Machines (SVM) have been studied. Various research work showed the success and superiority of ANN and SVM for load predictions, where frequently, SVM outperformed ANN models. In this study, data-driven thermal load forecasting performance of ε-SVM Regression (ε-SVM-R) based on a Radial Basis Function (RBF) and a polynomial kernel is compared to the outcome of two Nonlinear Autoregressive Exogenous Recurrent Neural Networks (NARX RNN) of different depths. For demonstration, historical data from a non-residential district in Germany is used for training and testing to predict monthly loads. The evaluation of the resulting predictions show that NARX RNNs yields higher accuracy than (ε-SVM-R) models, in combination with comparable computational effort.

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1. Introduction & state-of-the-art

The realisation of future sustainable energy systems is linked to the further development of district heating and cooling technologies, in order to increase energy efficiency, lowering costs and enable the implementation of smart energy systems. The 4th generation of district heating and cooling constitutes a key role to achieve these goals. An essential element in this context is the prediction of thermal loads [1,2].

Thermal load forecasting of buildings is mainly based on physical and data-driven models. Physical approaches require comprehensive information on thermal properties, materials or user behaviour [3]. This frequently results in parametrisation problems and has to be solved by a complex and costly data collection. However, taking data-driven approaches into consideration, detailed historical load profiles have to be available. In Neto et al. [4] the performance of an ANN was compared to the simulation outcome of a detailed physical model. The results were similar, although the Neural Network was trained with fewer parameters.

Statistical models forecast the current value of a variable by using previous time series values and previous or current values of exogenous factors such as weather and social variables. Stochastic time series, methods such as Autoregressive Moving Average (ARMA) or Autoregressive Integrated Moving Average (ARIMA) models appear to be among the most popular methods in short term load forecasting. Using a time series approach, a model is first developed based on the previous data before the future load is predicted based on this model. Fan and McDonald [5] applied stochastic time series techniques for load predictions in power systems. In multiple regression analyses, the statistical relationship between total load and weather conditions as well as other parameters that influence the load can be calculated. The regression coefficients are computed by an equally or exponentially weighted least squares estimation using the predefined amount of historical data. Dotzauer [6] proposed a prediction method based on a piecewise linear regression.

In recent times, research focused on approaches applying artificial intelligence. These techniques are based on processing big data and their learning abilities, including expert systems, neural networks or fuzzy neural models. However, artificial neural networks are the most commonly used methods, although, the model training takes usually a significant amount of time. Sakawa et al. [7] stated a prediction method based on a three-layered neural

* Corresponding author.
E-mail address: koschwitz@e3d.rwth-aachen.de (D. Koschwitz).

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network, with multiple output units. With respect to this algorithm, Kato et al. [8] proposed a backpropagation method and showed its superiority to the previous methods, whereas the work of Catalina et al. [9] takes numerous input parameters into detailed consideration based on a multiple regression model. Shamshirband et al. [10] constructed a neuro-fuzzy method to predict the heat load for individual consumers in a district heating system. In the context of network architectures as ANN representatives, NARX RNNs demonstrated promising performance in research for time series modelling. In Mena et al. [11] a NARX architecture is chosen to predict the electricity demand of a non-residential building. Powell et al. [12] tested different network architectures and showed a superior position of NARX for predicting electrical, heating and cooling loads. Various other studies proved its advantages for time series modelling and nonlinear systems [13,14]. Moreover, the capabilities of predicting chaotic times series with NARX RNNs were shown in Ref. [15]. Performance problems concerning long-term dependencies may also be reduced choosing this particular architecture [16]. Currently, studies on load forecasting strongly focus on SVM. For example, Li et al. [17] applied SVM to predict hourly cooling loads of an office building. Wu et al. [18] integrated a SVM model into a holistic building monitoring concept. In Magoulès and Zhao [19] different studies of building energy analysis are summarized, mainly focussing on forecast models. Based on the literature review, the authors applied a SVM model to predict electricity and heating load of several buildings in combination with an neural network for fault detection.

Amasyali et al. [20] provides a detailed literature review on building related data-driven load forecasting methods that have been studied so far. Numerous network architectures for ANN approaches have been tested. Frequently, the results have been compared to the outcome of SVM models, based on a RBF kernel. In most cases, both models obtained nearly the same performance level. In fact, SVM outperformed ANN approaches in many studies. Moreover, Amasyali et al. conclude that future research in this context should firstly focus on big energy data analytics, since larger quantities of monitoring data will be available in the future due to improved metering infrastructures and on the energy related behaviour of persons, which can be used to better understand their impact on the building-related overall energy consumption. Secondly, Deep Learning algorithms should be investigated due to their superior performance in comparison to traditional machine learning algorithms in other research areas [20].

The study presented in this work, addresses these issues in more detail. In this connection, the performance of a NARX RNN of varying depth is compared to a ε-SVM-R using a RBF and a polynomial kernel to predict thermal loads with the goal to provide progress for the current discussion on the superior performance of SVMs in comparison to ANNs. Adding more depth to the NARX RNN introduces a Deep Learning component that can be used to investigate possible performance improvements. In order to improve generalizing propositions, the test case includes detailed measurement data of 98 buildings within the heating system and measurement data of 47 buildings within the cooling system. Thus, possible differences between heating and cooling load prediction accuracy are investigated, which is motivated by different time series characteristics occurring frequently [12]. Moreover, the urban analyses level enables conclusions concerning big energy data analytics with respect to the applicability of different forecast models. In this context, model-related computational times are considered as well to be able to evaluate the computational effort of district forecast tasks.

The remainder of this paper is structured as follows: in section 2, the data set is statistically analysed and characterized. Subsequently, the data preprocessing is described and the ε-SVM-R models as well as the configurations of the NARX RNNs are presented. Section 3 comprises the results and discussions on the underlying data set as well as the chosen model parameters. In section 4, the results of this study are summarized.

2. Methods

2.1. Dataset for the following case study

The non-residential district for demonstration is located in Germany and comprises about 200 building characterized by years of construction, that differ between 1918 and 2015. The data set comprises historic time series data from 01/01/2016 to 01/01/2018 with an hourly resolution. Table 1 shows the energy consumers within the district heating and cooling system, depending on building usage classes. In order to train and test, the data set has been split into two parts. The input vectors are based on the following climate and occupancy feature selection: dew point temperature [°C], mean wind direction [°], mean wind velocity [m/s], outdoor temperature [°C], precipitation intensity [mm/h], precipitation quantity [mm], relative humidity [%], school holiday time [−], working time schedule [−].

Since a different behaviour in energy consumption between working days, weekends and holidays can be expected, occupancy variables (*school holiday time* and *working time schedule*) are integrated in the feature set. In this context, occupancy is represented by value 1 and non-occupancy by value 0. Working time is assumed to be equal for all buildings between 8:00 AM and 18:00 PM, since no detailed information were available. Equivalently, the variable *school holiday time* equals zero for holiday time. With regard to the climate features, the available sensor data has been chosen to describe ambient influences on heating and cooling loads. Besides the outdoor temperature, which strongly affect heating consumptions [6], additional six climate features complete the input data set. Relative humidity and dew point temperature affect the penetration of moisture into the construction, which leads to user-based heating regulations due to thermal comfort issues to prevent mould formation, for example. Mean wind direction and mean wind velocity affect thermal energy consumption due to infiltration, concerning certain thermal zones. Precipitation intensity and quantity are considered to be proxys for the degree of cloud coverage, which affects solar heat gains within a building. Table 2 shows minimum, maximum and mean values of the features.

For evaluating the relationship between input features and the

Table 1
Number of district heating and cooling consumers depending on building usage classes.

Building Usage	Heating (# consumers)	Cooling (# consumers)
Office	26	6
Office & Laboratory	26	19
Office & Repairing	7	–
Office & Facility	2	1
Office & Experimental Hall	1	1
Office & Teaching	1	–
Laboratory	15	11
Experimental Hall	8	7
Warehouse	5	–
Repairing & Warehouse	1	–
Social	4	1
Facility	1	1
Facility & Experimental Hall	1	–
	98	47

Table 2
Minimum, maximum, mean and standard deviation of feature time series.

	Outdoor temperature [°C]	Dew point temperature [°C]
Min.	−7.5	−10.0
Mean	11.4	7.8
Std. dev.	7.2	5.8
Max.	35.9	24.2
	Mean wind direction [°]	Mean wind velocity [m/s]
Min.	0.0	0.0
Mean	189.2	2.1
Std. dev.	66.1	1.3
Max.	325.7	10.2
	Precipitation quantity [mm]	Precipitation intensity [mm/h]
Min.	0.0	0.0
Mean	0.6	0.1
Std. dev.	1.6	0.3
Max.	25.6	15.2
	School holiday time [−]	Working time schedule [−]
Min.	0.0	0.0
Mean	0.8	0.3
Std. dev.	0.5	0.4
Max.	1.0	1.0
	Relative humidity [%]	
Min.	30.9	
Mean	76.1	
Std. dev.	15.6	
Max.	100.0	

heating and cooling load as output, a linear correlation model, based on hourly time series data is used. Fig. 1 illustrates linear correlation coefficients between heating loads and the feature variables, with respect to minimum, maximum and mean values. With regard to the data set used, the correlation between heating load, dew point temperature and outdoor temperature is high, although not every heating load is affected by the latter, according to the low minimum values. As mean wind direction, relative humidity, school holiday time and working time schedules show secondary influence, the correlation coefficient between mean wind velocity, precipitation quantity, precipitation intensity and heating load is low. In particular, taking the linear relationship between working time schedule and heating load as well as mean wind direction and heating load into consideration, only a few coefficients indicate a linear correlation with respect to the maximum values.

However, the median values are below 0.1, which implies that there is almost no linear correlation present for these variables. Fig. 2 illustrates the correlation coefficients between cooling loads and the feature variables on district scale.

The results tending to be similar compared to the heating load correlation analysis, although the mean and median values are significantly lower. The outcome of the heating-related correlation analysis on district scale is very similar to the results, based on building usage classes. Taking the present cooling load data of the experimental halls into consideration, the correlation coefficients between cooling loads and climate variables are lower, in contrast to the results of the correlation analysis on district scale. Overall, the cooling-related correlation coefficients vary between building usage classes. Although some scores are very low and redundant to other features, those features still might be useful for machine learning problems [21].

Before processing the data, it is crucial to get an idea of the data characteristics, since real-world data contain noise and originate from various data sources [22]. Therefore, variance and standard deviations for each hourly resolved time series with hourly resolution have been computed. Tables 3 and 4 show the mean values for heating an cooling load, depending on building usage classes.

Some time series show high standard deviations, which are attributable to outliers [19]. Thus, smoothing the data by removing outliers is a common choice. For this study, an outlier detection algorithm has been applied, which is introduced in the following subsection. Still, the aim was, to keep the signal as “original” as possible, since no further information about outlier verification has been available. Thus, the boundarys for outlier identification have been set appropriately, in order to avoid strong smoothing.

2.2. Data preprocessing

Since outliers significantly influence the performance of data analysis in general and load forecasting in particular, outlier detection is crucial. Outlier or anomalies within data can be distinguished, according to global outliers, contextual outliers and collective outliers. Global outliers are data points that strongly differ from the remaining data set. If the deviation of a data point

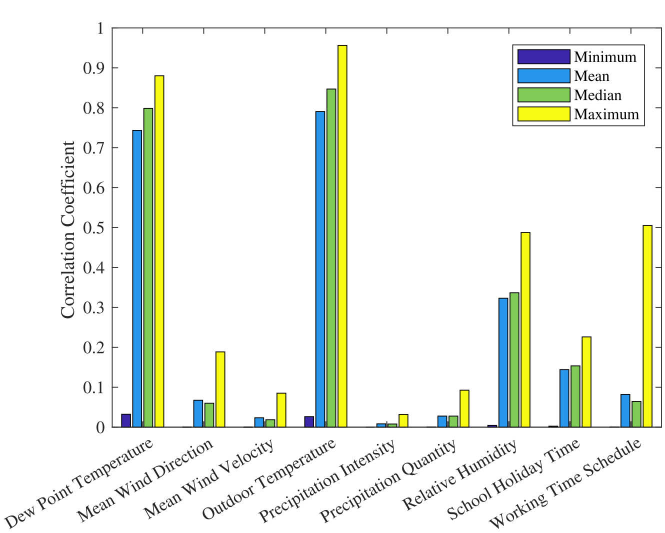

Fig. 1. Correlation coefficients for district heating loads to climate and occupancy features.

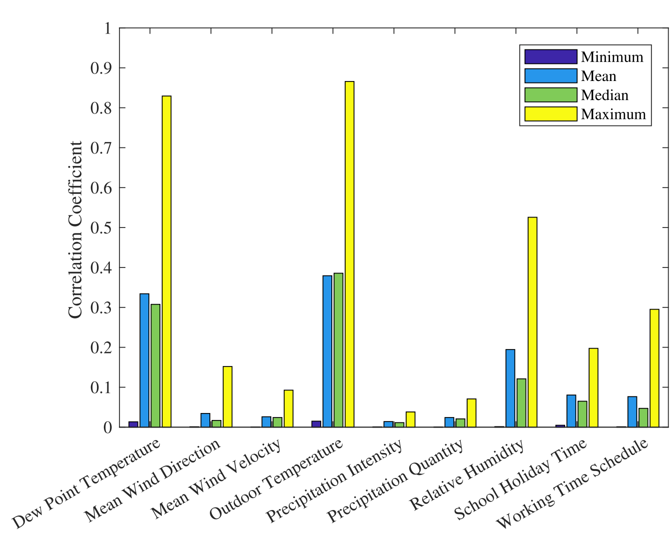

Fig. 2. Correlation coefficients for district cooling loads to climate and occupancy features.

Table 3
Mean and standard deviation of measured heating load time series depending on building usage classes.

Building Usage (# consumers)	μ [kW]	σ [kW]
Office (26)	94	66
Office & Laboratory (26)	103	79
Office & Repairing (7)	45	41
Office & Facility (2)	12	11
Office & Experimental Hall (1)	16	20
Office & Teaching (1)	33	29
Laboratory (15)	301	546
Experimental Hall (8)	91	77
Warehouse (5)	29	68
Repairing & Warehouse (1)	85	54
Social (4)	44	33
Facility (1)	453	278
Facility & Experimental Hall (1)	6	5

Table 4
Mean and standard deviation of measured cooling load time series depending on building usage classes.

Building Usage (# consumers)	μ [kW]	σ [kW]
Office (6)	84	44
Office & Laboratory (19)	194	390
Office & Facility (1)	198	48
Office & Experimental Hall (1)	23	83
Laboratory (11)	272	557
Experimental Hall (7)	476	65
Social (1)	29	11
Facility (1)	18	9

drifts based on the context like the measurement period, this data point can be declared a contextual outlier. A collective outlier is a group of data points that deviate from the remaining data set [22]. In research, various approaches for outlier detection are proposed. [23] contains a detailed survey of methodologies aiming at outlier detection. In this context, four main categories are presented: statistical methods, neural networks, machine learning techniques and hybrid systems. In Gupta et al. [24] outlier detection methods for temporal data has been studied, taking time series data, data streams, distributed data, spatio-temporal data and network data into consideration. For this study, a moving median method is applied to detect outliers. Therefore, a sliding window length of five time steps has been defined, where data points above or below three local median absolute deviations (MAD) are declared to be outliers. The MAD is defined according to equation (1). Those data points have been replaced by linear interpolated values.

$$MAD = median(|X_i - median(X)|) \tag{1}$$

Data scaling is a very important step within data preprocessing. Normalizing the input improves training performance and reduces the problem of outliers, since any average input, that is not close to zero, results in slowing down learning [25]. Subsequently, the input has been normalized between zero and one.

2.3. ε-support vector machine regression

The support vector representation was developed by Vladimir Vapnik in 1995. Based on this concept, Drucker et al. [26] introduced ε-SVM-R to represent continuous target values. The concept minimizes the regression error following the corresponding ε loss function L, with respect to equation (2), where y_i represents the target and F is the prediction function, depending on an input x_i, parameterized by weight $\widehat{w}$. [26]:

$$L = \begin{cases} 0 & \text{if } |y_i - F(x_i, \widehat{w})| < \varepsilon \\ |y_i - F(x_i, \widehat{w})| - \varepsilon & \text{otherwise} \end{cases} \tag{2}$$

The parameter ε defines a "tube" around the regression curve (see Fig. 3).

Predicted values that are within the ε-tube cause a loss of zero. For predicticted values outside the tube, the loss results from the difference between the value and the ε-tube [26]. Fig. 4 illustrates the ε-insensitive error function.

In order to solve this optimization problem, the error function has to be minimized based on equation (3), where U is a function of the slack variables ξ and ξ^*

$$U\left(\sum_{i=1}^{N}\xi_i^* + \sum_{i=1}^{N}\xi_i\right) + \frac{1}{2}\left(w^T w\right) \tag{3}$$

with respect to the following constraints, where b represents some bias and v_i represents the training instances.

$$\begin{aligned} & y_i - \left(w^T v_i\right) - b \le \varepsilon + \xi_i \\ & \left(w^T v_i\right) + b - y_i \le \varepsilon + \xi_i^* \\ & \xi_i^* \ge 0 \\ & \xi_i \ge 0 \end{aligned}$$

Since data points are frequently inside the tube, they are allowed to be on the "wrong" side of the margin, although with penalty. Therefore, ξ and ξ^* are defined. Introducing the Lagrange multipliers γ_i and α_i the Langrangian can be formulated by equation (4) [26].

$$\begin{aligned} L = {} & \frac{1}{2}\left(w^T w\right) + U\left(\sum_{i=1}^{N}\xi_i^* + \sum_{i=1}^{N}\xi_i\right) \\ & - \sum_{i=1}^{N}\alpha_i^*\left[y_i - \left(w^T v_i\right) - b + \varepsilon + \xi_i^*\right] \\ & - \sum_{i=1}^{N}\alpha_i\left[\left(w^T v_i\right) + b - y_i + \varepsilon + \xi_i\right] \\ & - \sum_{i=1}^{N}\left(\gamma_i^*\xi_i^* + \gamma_i\xi_i\right) \end{aligned} \tag{4}$$

Differentiating with respect to w_i, b and ξ leads to the equivalent dual space maximization problem, which is computationally simpler for solving the optimization problem. The corresponding objective function is given by equation (5), where P represents the polynomial degree:

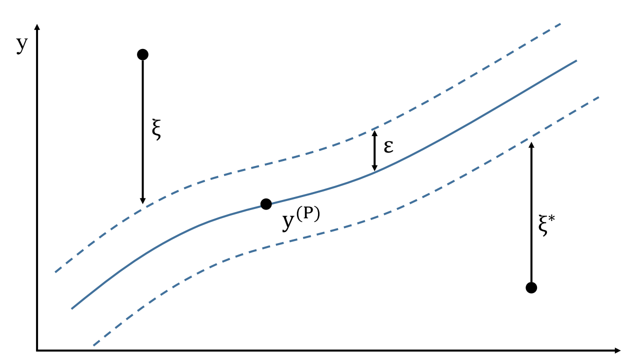

Fig. 3. Regression curve and ε-tube (based on [27]).

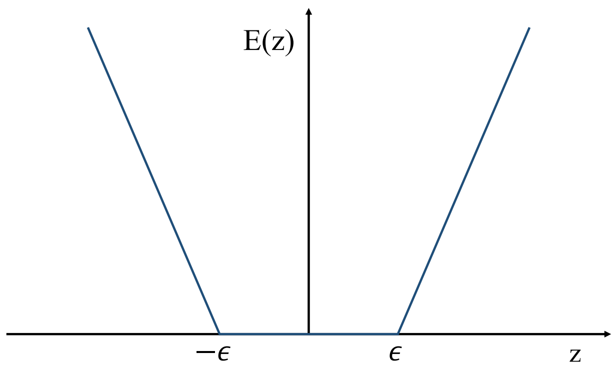

Fig. 4. Error function of ε-SVR (based on [26]).

$$W(\alpha,\alpha^*) = -\varepsilon\sum_{i=1}^{N}(\alpha_i^*+\alpha_i)+\sum_{i=1}^{N}y_i(\alpha_i^*-\alpha_i) \\ -\frac{1}{2}\sum_{i,j=1}^{N}(\alpha_i^*-\alpha_i)\left(\alpha_j^*-\alpha_j\right)\left(v_i^T v_j+1\right)^P \tag{5}$$

taking the following constraints into consideration

$$0 \le \alpha_i \le U \\ 0 \le \alpha_i^* \le U \\ \sum_{i=1}^{N}\alpha_i^* = \sum_{i=1}^{N}\alpha_i$$

Introducing a new vector $x^{(P)}$, the predicted value $y^{(P)}$ results from:

$$y^{(p)} = \sum_{i=1}^{N}(\alpha_i^*-\alpha_i)\left(v_i^T x^{(P)}+1\right)^P + b \tag{6}$$

fullfilling the Karush-Kuhn-Tucker (KKT) conditions [27]. Since regression problems cannot always be solved by linear models, kernel functions are introduced. Therefore, the dual formulation allows the integration of nonlinear functions in order to create a nonlinear regression model. The concept of kernel functions was introduced by Aizerman et al. (1964) and later used in the context of large margin classifiers [28]. The RBF is formulated in equation (7) and the polynomial kernel is defined by equation (8) [28].

$$K(x,x') = \exp\left(-\|x-x'\|^2\right) \tag{7}$$

$$K(x,x') = (x \bullet x' + 1)^P \tag{8}$$

In order to improve the choice for a certain kernel function, curve fittings serve as proxys. For this case study, a gaussian kernel has been chosen, since the tested curve fittings showed accurate results, as illustrated by Fig. 5.

To vary curve characteristics, a third-degree polynomial kernel has been applied. Fig. 6 shows a less accurate fit, compared to Fig. 5. Although, load profiles have been expected to be trained well by a ε-SVM-R, based on a third-degree polynomial kernel.

2.4. NARX Recurrent Neural Network

In this case study, two network architectures are chosen: A single hidden layer network structure and a two hidden-layered architecture. Each hidden layer comprises 30 hidden neurons. For

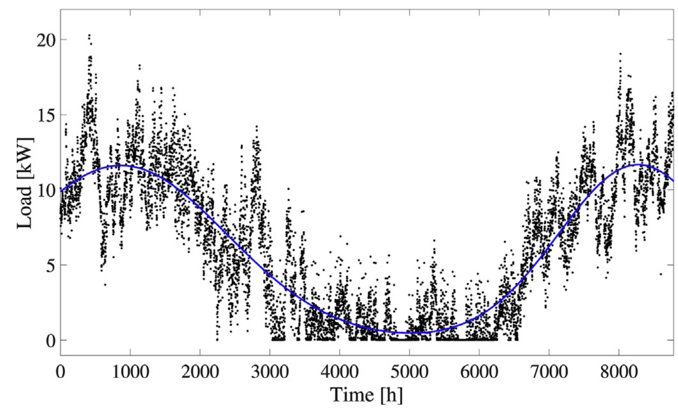

Fig. 5. Gaussian curve fitting.

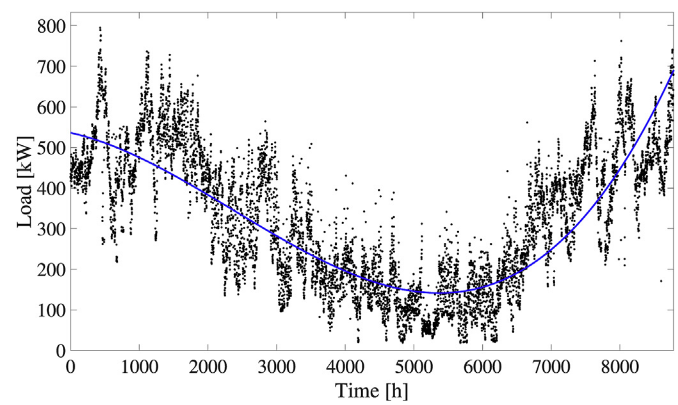

Fig. 6. Third-degree polynomial curve fitting.

optimization, the Levenberg-Marquardt algorithm is applied in order to update weights and bias [29,30]. Alternatively, Bayesian Regularization has been tested since this optimization algorithm improves the generalization property of the network [31]. In order to evaluate the performance of Levenberg-Marquardt and Bayesian Regularization, for this study, test predictions have shown, that by applying Bayesian Regularization, an improvement in accuracy of approximately 4% could be achieved. However, the computation time has been 23 times higher, compared to an optimization based on the Levenberg-Marquardt algorithm. Taking Bayesian Regularization into consideration, the Hessian of the second order partial derivatives has to be computed to define the curvature of a function and compute the gradient. However, the Levenberg-Marquardt optimization approximates the Hessian, which is computational less expensive, but also less accurate. Therefore, this algorithm usually provides the fastest convergence, which is considered to be crucial for load predictions on district scale in order to limit computational time. NARX RNNs are capable of embedding memory, which helps to deal with vanishing gradients. Therefore, several approaches are possible to model this characteristic. One opportunity is, to integrate time delays into neurons or connections between neurons [15]. In particular, firstly, a number of m time-delayed samples of the time series measurements $y(t)$ is defined, which secondly leads to the formulation of a reconstruction-space vector $\vec{R}(t)$, where τ represents the delay parameter [32].

$$\vec{R}(t) = [y(t), y(t-\tau), \ldots, y(t-(m-1)\tau)] \tag{9}$$

Garland and Bradley [33] show that high reconstruction dimensions do not necessarily improve forecast accuracy of noisy time series, since every past noisy data point influence the predicted data point. In this study, an input and a network feedback delay of one time step is integrated. Fig. 7 illustrates the chosen recurrent network architecture with one hidden layer, where z^{-1} represents the time delays.

For both network architectures, sigmoidal activation functions are used. Due to their continuous functional characteristics, this kind of activation function is well suited for time series forecasts. The saddle points at zero and one moreover enables to model minimum or maximum load behaviour [12]. Equation (10) represents the activation functions ϕ of each hidden neuron j, depending on the input u.

$$\phi_j(u_i) = \tanh\left(\sum_{i=1}^{M} w_{i,j} u_i + b_i\right) \tag{10}$$

Adding the weights w for each connection between input neuron and hidden neuron as well as some bias b for each hidden neuron, the activation function is computed. Equation (11) represents the sum of outputs from the hidden layer, with respect to the weights corresponding to the connection between hidden neurons and output neuron W and bias B.

$$\Omega(\phi_j) = \sum_{j=1}^{N} W_j \phi_j + B \tag{11}$$

The output is fed back to the input of the network, which defines the recursive characteristics. Taking the training procedure into consideration, a learning automata approach of Narendra and Parthasarathy is applied [35]. The authors showed, that a hierarchical network architecture without feeding back the output to the input is well-suited for training, if the true output is available. Thus, the closed-loop form of the network for predictions is replaced by a open-loop architecture.

2.5. Performance indicators

For performance measurements, the Mean Average Error (MAE) and Mean Squared Error (MSE) have been chosen to characterize model performances. These measures can be calculated based on (12) and (13).

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_{predict,i} - y_{data,i}\right| \tag{12}$$

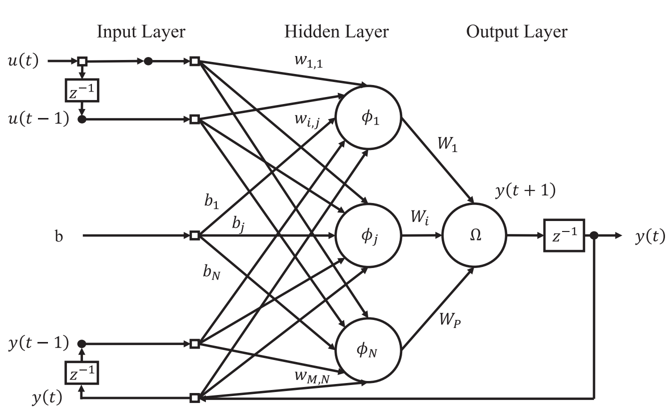

Fig. 7. Architecture of NARX Recurrent Neural Network with one hidden layer (own illustration based on [34]).

$$MSE = \frac{1}{n}\sum_{i=1}^{n}\left(y_{predict,i} - y_{data,i}\right)^2 \tag{13}$$

Compared to MAE, MSE is stronger influenced by large errors, since the errors increase by a power of two [36].

3. Results

In Table 5, the median of MAE and MSE values for heat load predictions are shown, depending on the forecast model and season. Table 6 illustrates the results for MAE and MSE, based on cooling load predictions.

Taking these performance indicators into consideration, the NARX RNN configurations show better performance, compared to the ε-SVM-R models for heating and cooling load predictions, which is exemplary illustrated by Fig. 8.

The latter shows a 40-h excerpt of predicted cooling loads in

Table 5
Overview of median MSE and MAE values based on heating load predictions depending on different seasons.

Season	NARX RNN 1 hidden Layer	NARX RNN 2 hidden Layer
Winter	MAE: 3,82 MSE: 52,45	MAE: 4,09 MSE: 49,44
Spring	MAE: 3,73 MSE: 50,33	MAE: 3,79 MSE: 52,22
Summer	MAE: 3,92 MSE: 49,00	MAE: 4,12 MSE: 50,10
Autumn	MAE: 3,89 MSE: 52,20	MAE: 4,01 MSE: 51,52
	ε-SVM-R RBF Kernel	**ε-SVM-R polynomial Kernel**
Winter	MAE: 16,38 MSE: 419,87	MAE: 13,03 MSE: 263,45
Spring	MAE: 9,46 MSE: 159,68	MAE: 8,86 MSE: 151,86
Summer	MAE: 7,35 MSE: 158,90	MAE: 9,33 MSE: 120,00
Autumn	MAE: 9,66 MSE: 163,55	MAE: 10,30 MSE: 181,58

Table 6
Overview of median MSE and MAE values based on cooling load predictions depending on different seasons.

Season	NARX RNN 1 hidden Layer	NARX RNN 2 hidden Layer
Winter	MAE: 2,44 MSE: 27,46	MAE: 2,57 MSE: 28,54
Spring	MAE: 2,39 MSE: 26,95	MAE: 2,41 MSE: 33,20
Summer	MAE: 2,39 MSE: 41,72	MAE: 2,54 MSE: 30,28
Autumn	MAE: 2,35 MSE: 30,16	MAE: 2,85 MSE: 40,40
	ε-SVM-R RBF Kernel	**ε-SVM-R polynomial Kernel**
Winter	MAE: 7,77 MSE: 80,70	MAE: 6,60 MSE: 71,80
Spring	MAE: 7,97 MSE: 90,55	MAE: 5,82 MSE: 69,57
Summer	MAE: 14,66 MSE: 246,23	MAE: 13,94 MSE: 259,96
Autumn	MAE: 9,12 MSE: 140,90	MAE: 9,37 MSE: 131,98

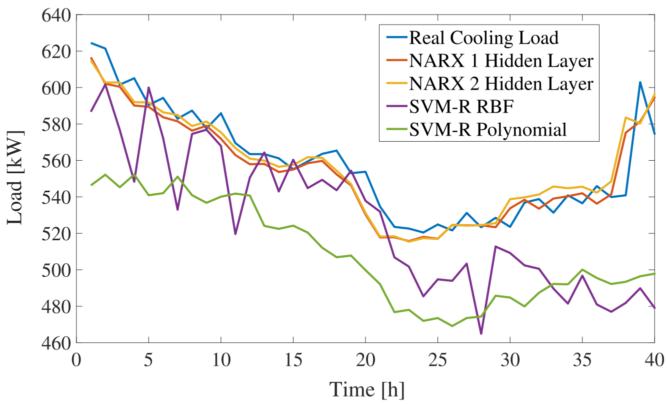

Fig. 8. Example for cooling load predictions.

July. The MAE values of NARX RNN with one hidden layer and the configuration based on two hidden layers are very similar, although the first network shows a slightly better performance. Regarding the median of MSE, in some cases the two hidden layered network perform better. The results also indicate that the NARX RNNs perform nearly constant during seasons. Based on RBF and polynomial kernel, the seasonal performance of the ε-SVM-R models is inconsistent. In terms of heating load predictions, the RBF model shows a lower MAE value in summer and autumn time compared to the polynomial model, whereas in winter and spring the polynomial model performs better on MAE. Except for predictions in autumn, ε-SVM-R with polynomial kernel shows a lower MSE value. The winter forecasts with the ε-SVM-R models show higher errors, compared to the remaining seasonal prediction results. One reason could be the varying user behaviour due to individual thermal comfort and ventilation habits, which is not represented in sufficient detail by the features. As NARX RNNs take historical time series elements including the real user behaviour within the measured values into account, the good performance of the neural networks in winter time might be explained.

The results of cooling load predictions show a better MAE performance of the polynomial model in winter, spring and summer. Additionally, the MAE values in autumn are very similar. The highest median MAE and MSE based on heating load predictions occur during heating season in winter. Equivalently, the highest median MAE and MSE based on cooling load predictions occur during cooling season in summer. It is noticable, that the ε-SVM-R models showed its worst performance in summertime, which can be again attributed to the user behaviour in reversed analogy to the heating load predictions.

In order to enable a more differentiated view on the results, MAE values for each time series forecast provide a basis for further performance investigations. Tables 7 and 8 list the number of minimum MAE values, based on a model-related comparision.

Therefore, for each time series prediction task, the resulting MAE values, induced by the four different forecast models, form the basis to compute the minimum MAE value for each prediction task. Subsequently, the model providing the best performance has been identified. Repeating this procedure for each prediction task, the number of best performances has been calculated. The results indicate, that NARX RNNs outperformed the ε-SVM-R in most cases, with respect to heating load predictions. Considering cooling load predictions, the results are slightly different. NARX RNNs show higher accuracy in most prediction cases, though the ε-SVM-R models frequently demonstrate better performance. However, some predictions, based on ε-SVM-R are of comparable accuracy, as shown by Fig. 9.

Table 7
[Number of minimum MAE values based on seasonal heating load predictions.

Season	NARX RNN 1 hidden Layer	NARX RNN 2 hidden Layer
Winter	54	41
Spring	63	31
Summer	56	31
Autumn	54	36
	ε-SVM-R RBF Kernel	**ε-SVM-R polynomial Kernel**
Winter	0	2
Spring	2	1
Summer	6	4
Autumn	3	4

Table 8
Number of minimum MAE values based on seasonal cooling load predictions.

Season	NARX RNN 1 hidden Layer	NARX RNN 2 hidden Layer
Winter	19	13
Spring	25	10
Summer	24	11
Autumn	22	15
	ε-SVM-R RBF Kernel	**ε-SVM-R polynomial Kernel**
Winter	12	5
Spring	5	9
Summer	5	9
Autumn	4	8

A reason for the lower performance level of the RNNs concerning cooling load predictions are due to the weather-independent process cooling and the human behaviour of switching the chillers on and off at various times. Fig. 10 shows an exemplary cooling load time series. Since the RNN models feed the output back to the input, the prediction of a further time step results in a high error due to an user intervention at the current time step. In this case, the network tends to overfit.

As computational effort affects the model selection, the computational time for predictions, including training and testing, has been measured. In this study, the application of the ε-SVM-R model based on RBF required about 7 s on average for training and testing for each time series. Using the polynomial kernel, the ε-SVM-R model provides the result after about 358 s on average. The NARX RNN with one hidden layer required about 13 s and the

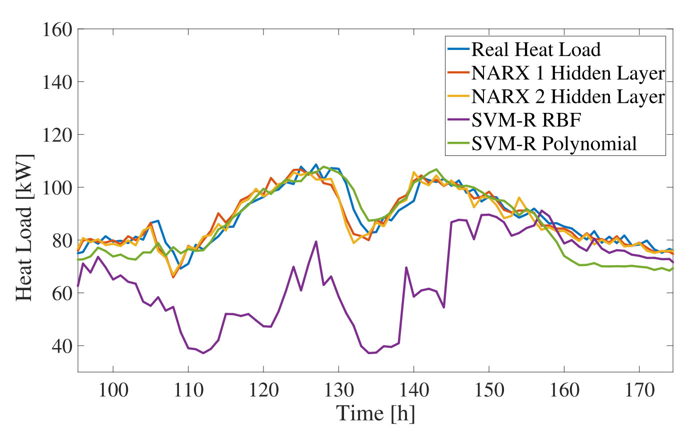

Fig. 9. Example for resulting heating load prediction.

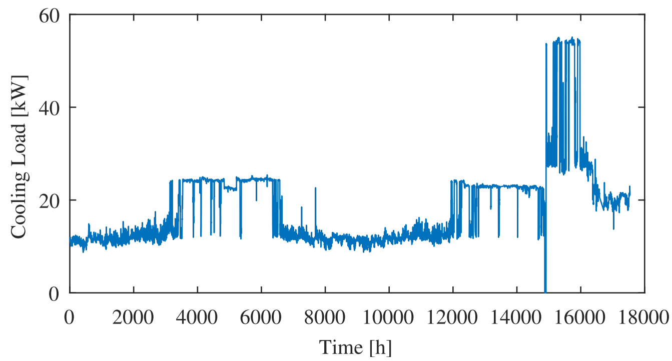

Fig. 10. Cooling load time series.

NARX configuration with two hidden layer about 57 s on average for each prediction. Taking the computational effort and the error measurements into account, the NARX RNN with one hidden layer provides a considerable overall performance.

4. Conclusion

In this district scale case study, the thermal load forecasting performance of ε-SVM-R models has been compared to the outcome of two NARX RNNs of different depth. Taking the results into consideration, the NARX RNNs outperformed the ε-SVM-R models based on the MAE and MSE. Although, the neural networks tend to overfit in case of user-driven cooling load time series due to their recursive abilities. The lowest accuracy concerning heating and cooling load predictions based ε-SVM-R models occur in winter respectively in summertime, which can be explained by human behaviour based on the individual thermal comfort. This study also shows, that deeper network architectures not necessarily lead to better results in data-driven thermal load predictions. The performance of NARX RNNs are strongly affected by the dimension embedded memory of input and ouput and the depth of the network architecture, since long-term dependencies may result in vanishing gradients [15]. Moreover, Bayesian Regularization has been tested for optimization to possibly replace the Levenberg-Marquardt algorithm, since results frequently provide higher accuracy. Although, increasing performance has been proven in the case of this study, computation time has to be considered for prediction tasks on urban scale. The results have to be assessed against the background of the underlying data set. As shown by statistical analyses, the measured time series are noisy, which affect the prediction quality. Furthermore, correlation analyses indicates, that the available data might not be sufficient, especially concerning feature representations for cooling load predictions. Still, the NARX RNNs performed well despite the building-related information insufficiency, since feeding back measured time series data already contain user behaviour. Taking the computational effort into account, the NARX RNN with one hidden layer requires nearly the same time for training and testing compared to the ε-SVM-R, based on RBF. Deeper network architecture of NARX RNN leads to computational costs that are four times higher than for a comparable single-layered NARX RNN.

It could be shown that NARX RNNs are very powerful and well-suited for predicting tasks related to district heating and cooling systems. Here, they outperformed the ε-SVM-Rs, introduced in this work, with respect to prediction accuracy. This, however, contrasts current findings from the literature, which state that SVMs outperform ANN models in general [19,20]. Furthermore, this work indicates that in particular cases deeper NARX RNNs are able to improve forecast results of a building's energy demand. The majority of the investigated single buildings, however, showed better prediction accuracy for single-layered NARX RNNs (see Table 6). On the urban scale, which is represented as the aggregated results of the single building's energy predictions, the load forecast accuracy improved as well. Even though, it might be possible that deep NARX RNNs show better performance regarding load forecasts, when based on aggregated input data representing an entire district, which has to be investigated in future research work. Since computational effort is highly relevant, especially for big energy data analytics, model-related computational time has been analyzed with respect to different optimization algorithms and model configurations.

Overall, on the basis of this study, NARX RNN configurations with one hidden layer are recommended for comparable use cases due to their high forecast accuracy and low computational effort. Further research work concerning data-driven cooling load predictions is required to investigate the overfitting tendency of NARX RNNs related to individual load profile characteristics of cooling load time series or varying circumstances such as changes of technical equipment or influences of district characteristics linked to different building types.

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